0580_w23_qp_32_Q8

8 **(a)** In triangle RST, $RT = 7$ cm and $ST = 4$ cm.

(i) **Using a ruler and compasses only**, construct triangle RST.
Leave in your construction arcs.
The line RS has been drawn for you.

[2]

(ii) Measure the distance from S to the midpoint of RT.
Give your answer in millimetres.

..... mm [1]

(b) Town A is 8.5 cm from town B on a map.
The scale of the map is 1 : 50 000.

Calculate the actual distance from town A to town B.
Give your answer in kilometres.

..... km [2]

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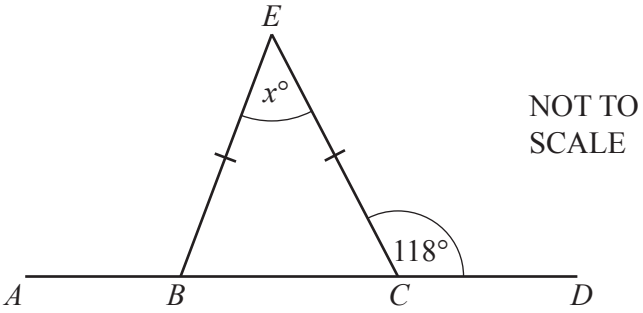
(c)

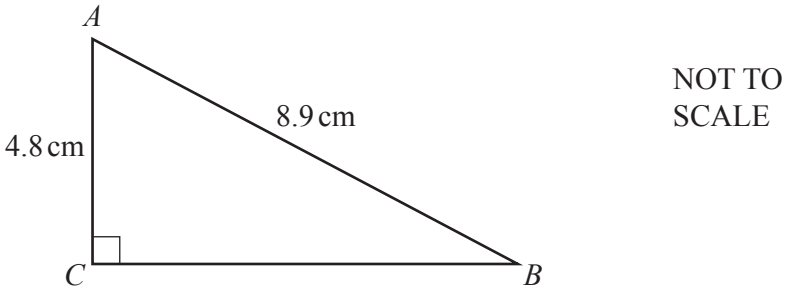


The diagram shows triangle *BCE* and a straight line *ABCD*.
BE = *CE* and angle *DCE* = 118°.

Find the value of *x*.

x = ... [2]

(d)



The diagram shows a right-angled triangle *ABC*.

Show that *BC* is 7.5 cm, correct to 2 significant figures.

[3]